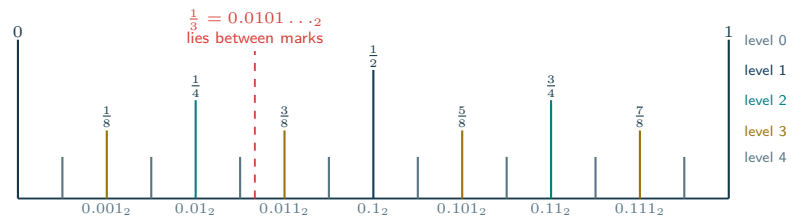
ARITHMETIC ON ONE PAGE

Dyadic numbers: a ring that inverts only 2

Fractions with power-of-two denominators form a ring, and removing the prime 2 splits one factor off $\zeta(2)$ and $\xi(2)$.

1 / THE RULE AND THE PICTURE

A dyadic rational is $m/2^k$, and the ruler adds the odd multiples of $1/2^k$ at level k



Level k of the ruler adds the 2^{k-1} marks with an odd top, and $2^k - 1$ marks exist after k levels. These are exactly the numbers whose binary expansion stops. Sums and products stay on the ruler,

$$\boxed{\mathbb{Z}[\tfrac{1}{2}] = \{\, m/2^k : \ m \in \mathbb{Z},\ k \geq 0 \,\}}$$

as in $\frac{7}{8} + \frac{5}{16} = \frac{19}{16}$ and $\frac{1}{2} \cdot \frac{3}{8} = \frac{3}{16}$, but dividing can leave it: $\frac{1}{2} \Big/ \frac{3}{2} = \frac{1}{3}$.

2 / WHAT CAN BE DIVIDED BY

Exactly the powers of two, because only 2 has been inverted

ring	numbers it can divide by	primes inverted
$\mathbb{Z}$	± 1 only	none inverted
$\mathbb{Z}[\frac{1}{2}]$, the dyadic rationals	$\pm 2^k$ for every whole k	2
$\mathbb{Z}[\frac{1}{6}]$	$\pm 2^a 3^b$	$2,\ 3$
$\mathbb{Q}$, a field	every nonzero number	all primes

A dyadic number $m/2^k$ with m odd has a dyadic reciprocal only when $m = \pm 1$, so $\frac{3}{8}$ cannot be divided by.
Writing any nonzero rational as $2^k \cdot a/b$ with a, b odd, it is dyadic exactly when $b = 1$, that is, when no odd prime divides the denominator. Inverting every prime gives the field $\mathbb{Q}$.

3 / SPLITTING THE INTEGERS BY POWERS OF TWO

Every $n = 2^k m$ with m odd, so $\zeta(2)$ splits into a dyadic part and an odd part

Summing $1/n^2$ over all whole numbers and splitting $n = 2^k m$ gives

$$\boxed{\zeta(2) = \left(\sum_{k \geq 0} \tfrac{1}{4^k}\right)\left(\sum_{m \text{ odd}} \tfrac{1}{m^2}\right) = \tfrac{4}{3} \cdot \tfrac{\pi^2}{8} = \tfrac{\pi^2}{6}}$$

The dyadic factor $\frac{4}{3} = \frac{1}{1-2^{-2}}$ is the Euler factor at 2, and what remains, $\frac{\pi^2}{8} = 1.2337$, is the sum over the odd whole numbers, which is what is left once 2 has been inverted. Each further prime p inverted removes a factor $\frac{1}{1-p^{-2}}$:

ring	primes inverted	sum of $1/m^2$ over whole m coprime to them	value
$\mathbb{Z}$	none	$\pi^2/6$	1.6449
$\mathbb{Z}[\frac{1}{2}]$	2 (factor $\frac{4}{3}$)	$\pi^2/8$	1.2337
$\mathbb{Z}[\frac{1}{6}]$	$2, 3$ (also $\frac{9}{8}$)	$\pi^2/9$	1.0966
$\mathbb{Z}[\frac{1}{30}]$	$2, 3, 5$ (also $\frac{25}{24}$)	$(\pi^2/6) \cdot \frac{3}{4} \cdot \frac{8}{9} \cdot \frac{24}{25}$	1.0528
$\mathbb{Q}$	every prime	1	1

4 / THE COMPLETED VALUE $\xi(2)$

The real line supplies the factor $1/\pi$, and the primes supply the rest

The completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ equals $\xi(2) = \pi^{-1}\zeta(2) = \frac{\pi}{6} \approx 0.5236$, the angle $30°$ in radians.
In product form $\xi(2) = \frac{1}{\pi} \cdot \frac{4}{3} \cdot \frac{9}{8} \cdot \frac{25}{24} \cdots$, with $\frac{1}{\pi}$ from the real line, where the ruler's dyadic marks are dense, and $\frac{4}{3}$ from the prime 2. The functional equation $\xi(s) = \xi(1-s)$ gives $\xi(-1) = \frac{\pi}{6}$ as well, and since $\xi(-1) = \sqrt{\pi}\,\Gamma(-\frac{1}{2})\zeta(-1) = -2\pi\,\zeta(-1)$ this forces $\zeta(-1) = -\frac{1}{12}$.
One number $\frac{\pi}{6}$ therefore holds both $\zeta(2) = \frac{\pi^2}{6}$ and $\zeta(-1) = -\frac{1}{12}$.

ASIDE / TRY THESE

Is $\frac{5}{12}$ dyadic? No, $12 = 4 \cdot 3$. The odd sum $1 + \frac{1}{9} + \frac{1}{25} + \cdots$ is about 1.2337.

LINKS TO OTHER SHEETS

The adding-a-new-number sheet gives ring and field; the Basel sheet proves $\zeta(2) = \frac{\pi^2}{6}$.

Convention. $\mathbb{Z}[\frac{1}{2}]$ the dyadic rationals; $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, with $\xi(2) = \xi(-1) = \frac{\pi}{6}$; ring sums were checked numerically.

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